RESEARCH PAPER

Customer Churn Prediction in Subscription-Based Services:

An End-to-End Analysis Using XGBoost, SHAP Explainability, and Agentic AI Perspectives

Mohammed Amine

Casablanca, Morocco

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ABSTRACT

Customer churn remains a critical challenge for subscription-based businesses, including telecommunications and SaaS providers, where retaining existing customers is often more cost-effective than acquiring new ones. This study presents a comprehensive end-to-end analysis of the IBM Telco Customer Churn dataset comprising 7,043 records. Through exploratory data analysis, feature engineering, predictive modeling with XGBoost, and interpretability via SHAP (SHapley Additive exPlanations) values, key churn drivers were identified — including contract type, tenure, monthly charges, and internet service type. The model achieved a ROC-AUC score of 0.8389 with an overall accuracy of approximately 79%. A business impact simulation quantified potential revenue leakage and estimated annual savings of up to $40,320 USD under a 15% relative churn reduction scenario. Finally, the analysis incorporates a forward-looking discussion on agentic AI systems for proactive, autonomous retention strategies. The findings provide actionable insights for reducing churn and enhancing customer lifetime value in competitive subscription markets.

1. Introduction

In competitive subscription markets, customer churn directly erodes recurring revenue and increases reliance on costly acquisition channels. Effective churn prediction and mitigation strategies are essential for sustainable business growth, particularly in telecommunications, where contract flexibility, service quality, and pricing sensitivity significantly influence customer retention decisions.

The economic rationale for investing in churn prevention is compelling: acquiring a new customer typically costs five to seven times more than retaining an existing one, and even small improvements in retention rates can yield disproportionately large gains in profitability and customer lifetime value (CLV). In this context, data-driven approaches to churn prediction have become a cornerstone of modern customer relationship management (CRM).

This paper conducts a systematic, end-to-end analysis of the publicly available IBM Telco Customer Churn dataset to uncover churn patterns, develop a robust predictive model, quantify the associated financial impact, and explore emerging technological solutions. The study is structured around five primary objectives:

- To identify the main demographic, behavioral, and service-related drivers of churn through exploratory data analysis (EDA) and feature engineering.

- To construct and evaluate a machine learning model — specifically an XGBoost classifier — capable of accurately predicting customer churn.
- To apply explainable AI (XAI) techniques, specifically SHAP values, for model interpretability and the derivation of actionable insights.
- To simulate the business impact of churn reduction under realistic intervention scenarios.
- To discuss the transformative role of agentic AI in shifting retention strategies from reactive to autonomous, real-time interventions.

By integrating rigorous statistical analysis, gradient boosting models, and emerging AI paradigms, this study demonstrates how data science can deliver measurable business value in competitive subscription-based industries. The findings are particularly relevant to the Moroccan telecommunications sector, where increasing competition and evolving consumer expectations make intelligent retention strategies a critical differentiator.

2. Data and Methodology

2.1 Dataset Description

This analysis utilizes the IBM Telco Customer Churn dataset, a widely used benchmark in churn prediction research. The dataset comprises 7,043 customer records across 21 attributes, providing a comprehensive view of customer demographics, account characteristics, service subscriptions, and billing information.

The variables span three categories: (i) demographic information — gender, SeniorCitizen status, presence of Partner and Dependents; (ii) account details — tenure (months with the company), Contract type, and PaymentMethod; and (iii) service subscriptions — PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies, alongside billing variables MonthlyCharges and TotalCharges.

The binary target variable **Churn** indicates whether a customer departed within the observation period (Yes/No). The dataset exhibits moderate class imbalance, with an observed churn rate of **26.54%** (1,869 churned customers out of 7,043), necessitating careful model evaluation beyond simple accuracy metrics.

2.2 Tools and Technologies

All analyses were performed in a Python environment using Google Colab. The complete analytical stack comprised:

- **Data manipulation:** pandas and numpy for data wrangling, transformation, and feature engineering.
- **Visualization:** matplotlib, seaborn for static plots, and plotly for interactive exploratory charts.
- **Machine learning:** scikit-learn for preprocessing pipelines and evaluation, xgboost for gradient boosted classification.
- **Explainability:** shap for global and local model interpretability via SHapley Additive exPlanations.
- **Preprocessing utilities:** ColumnTransformer, StandardScaler, and OneHotEncoder from scikit-learn.

2.3 Data Preparation and Feature Engineering

Initial preprocessing addressed data quality issues, most notably the TotalCharges column, which was stored as an object type due to whitespace entries for new customers with zero tenure. These were converted to numeric, with missing values imputed as zero. The customerID field was dropped as it carries no predictive signal.

Four engineered features were created to enhance model performance and interpretability:

- **AvgMonthlyCharge:** Computed as TotalCharges / tenure (with a safeguard against zero-division for new customers). This feature captures the effective average spend per active month and is more informative than raw TotalCharges for customers of varying tenures.
- **TotalServices:** A count of all subscribed value-added services (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies). Higher engagement breadth may signal lower churn risk.
- **HasFiber:** A binary indicator (0/1) for Fiber optic internet service, motivated by the markedly higher churn rate observed in this segment.
- **ContractType:** A simplified categorical encoding of contract duration into a three-level ordinal variable (Monthly, One-Year, Two-Year), designed to capture the strong contractual lock-in effect on churn.

These features capture customer engagement intensity, service complexity, and contractual commitment — all theoretically and empirically linked to churn risk in subscription markets.

2.4 Exploratory Data Analysis

Comprehensive univariate and bivariate analyses were conducted to understand the distribution of variables and their relationships with churn. Visualizations included bar charts for categorical distributions stratified by churn status, box plots and histograms for continuous variables

(MonthlyCharges, tenure), correlation heatmaps for numeric features, and interactive Plotly charts for dynamic exploration.

Key patterns observed during EDA informed both feature engineering decisions and the framing of business recommendations, and are discussed in detail in Section 3.

3. Exploratory Data Analysis & Key Findings

3.1 Churn Rate Overview

The overall churn rate of **26.54%** indicates that more than one in four customers departed during the observation period — a significant proportion that underscores the business urgency of this analysis. The remaining 73.46% of customers were retained (No Churn).

3.2 Churn by Customer Segment

Bivariate analysis revealed substantial heterogeneity in churn rates across customer segments. The table below summarizes the most informative categorical breakdowns:

Segment	Sub-Group	Churn Rate (%)
Internet Service	Fiber Optic	41.89%
Internet Service	DSL	18.96%
Internet Service	No Internet	7.40%
Payment Method	Electronic Check	45.29%
Payment Method	Mailed Check	19.11%
Payment Method	Bank Transfer (Auto)	16.71%
Payment Method	Credit Card (Auto)	15.24%
Senior Citizen	Yes	41.68%
Senior Citizen	No	23.61%
Partner Status	No Partner	32.96%
Partner Status	Has Partner	19.66%
Dependents	No Dependents	31.28%
Dependents	Has Dependents	15.45%

Table 1. Churn rates across key customer segments (source: IBM Telco dataset analysis)

These results reveal several important patterns. Fiber optic internet subscribers churn at a rate more than twice that of DSL subscribers, suggesting potential value-perception gaps related to

pricing or service quality. Electronic check users exhibit the highest churn among payment methods at 45.29%, likely reflecting lower institutional commitment compared to automatic payment users. Senior citizens churn at nearly twice the rate of non-senior customers, and customers without partners or dependents show substantially higher churn — consistent with the hypothesis that social and familial anchoring reduces propensity to switch.

3.3 Churn by Contract Type

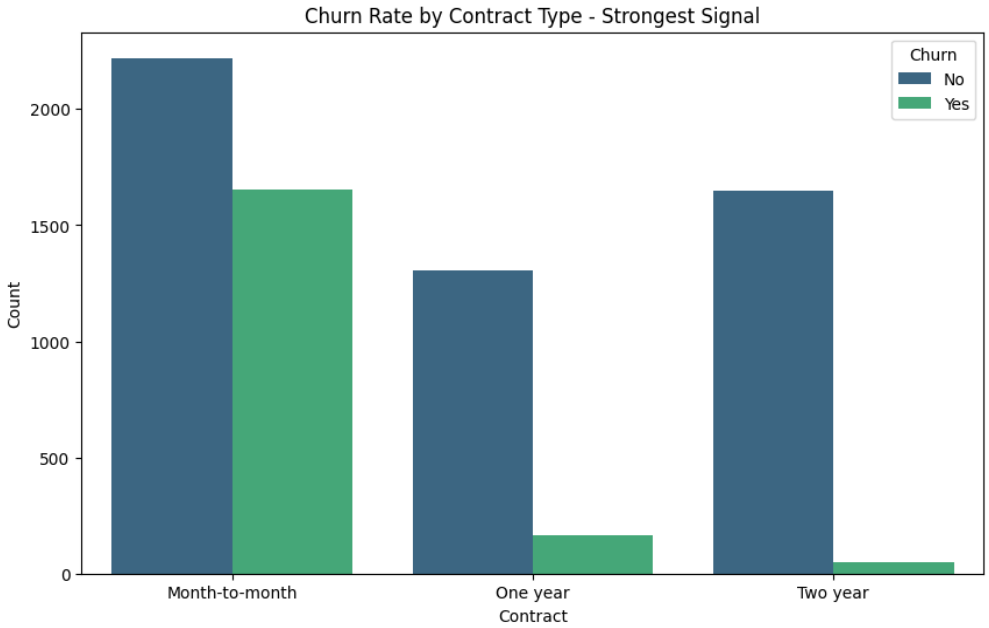

Figure 1. Churn rate by contract type — month-to-month contracts exhibit dramatically higher churn

Contract type emerges as the single most powerful predictor of churn. Month-to-month contract holders churn at a dramatically higher rate compared to one-year and two-year contract customers. This is consistent with the economic logic of switching costs and sunk cost effects: longer-term contracts create commitment mechanisms that substantially reduce churn probability. This finding directly motivates the recommendation to incentivize longer-term contracts at the point of customer acquisition.

3.4 Churn by Tenure

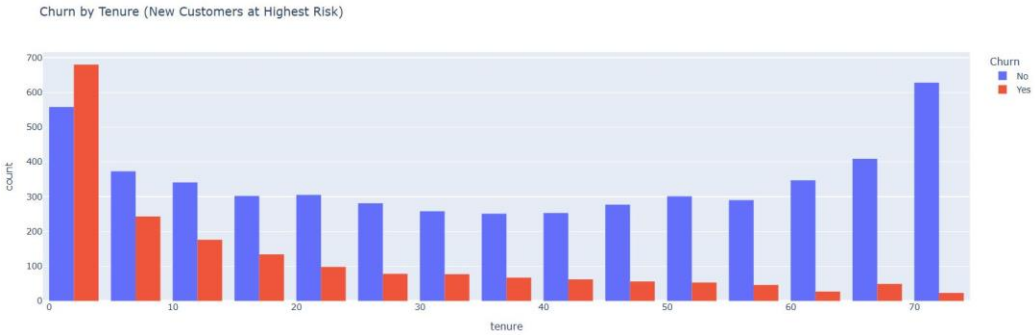

Figure 2. Churn rate by customer tenure — early-tenure customers are at highest risk

Tenure exhibits a clear inverse relationship with churn risk. Customers in their first six to twelve months of service are substantially more likely to churn than longer-tenured customers. This early-churn phenomenon is well-documented in subscription economics and motivates a disproportionate investment in early-customer-lifecycle interventions — onboarding quality, early value realization, and proactive check-ins during the critical first year.

3.5 Monthly Charges Distribution

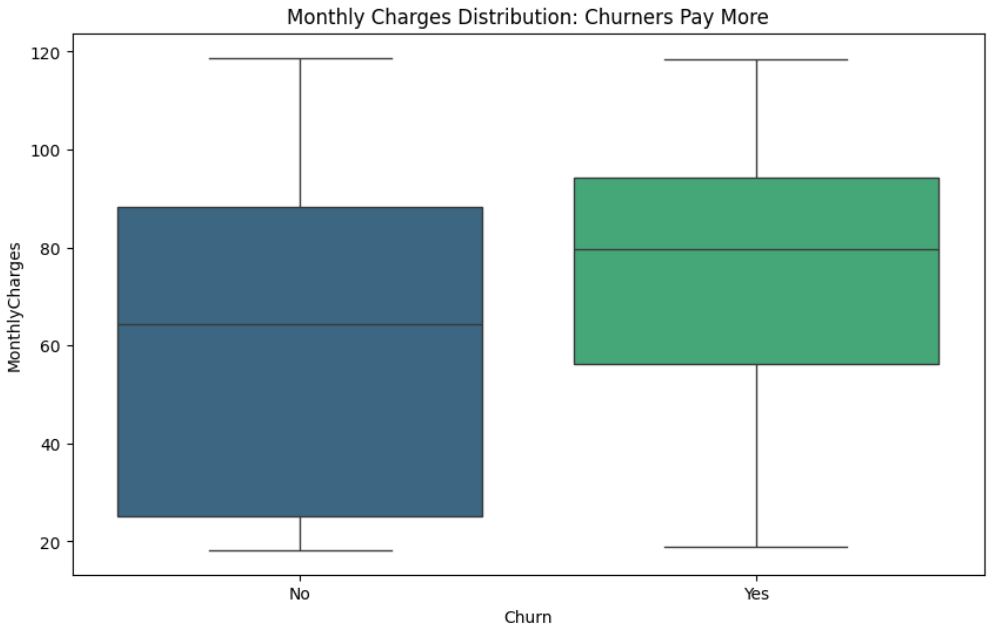

Figure 3. Distribution of monthly charges by churn status — churners tend to have higher monthly charges

Churned customers exhibit systematically higher monthly charges than retained customers. This pattern likely reflects a perceived value-to-price imbalance: customers paying premium rates without a commensurate perceived service value are more susceptible to competitive alternatives. This finding reinforces the importance of value communication and service quality assurance, particularly for high-spend customer segments.

3.6 Feature Correlation Analysis

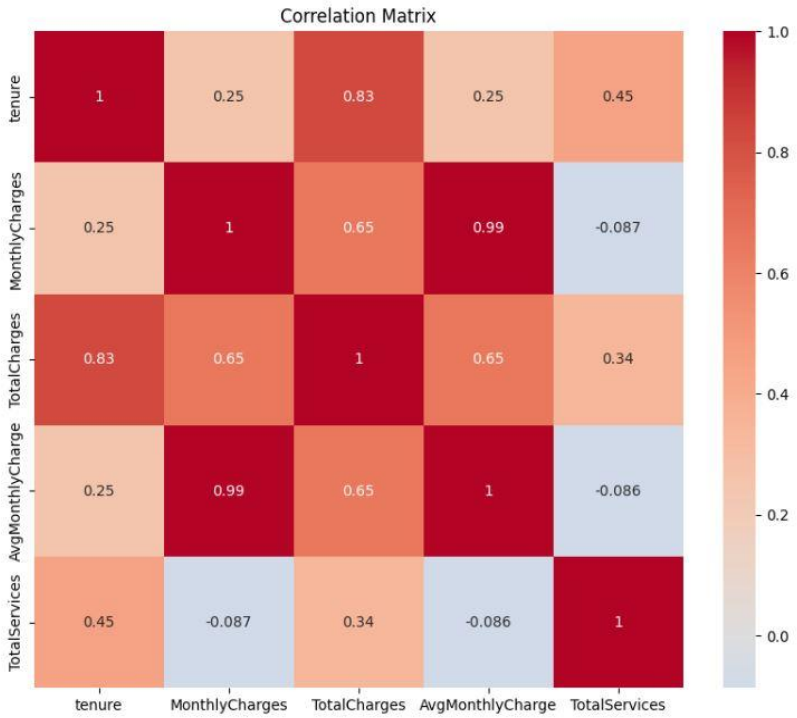

Figure 4. Correlation matrix of numeric features — tenure negatively correlates with churn while monthly charges correlate positively

The correlation heatmap confirms the directional relationships observed in bivariate analysis. Notably, tenure shows a negative correlation with churn, MonthlyCharges shows a positive correlation, and TotalCharges is strongly collinear with both tenure and MonthlyCharges, partially motivating the derivation of the AvgMonthlyCharge engineered feature to decorrelate these signals.

4. Predictive Modeling

4.1 Modeling Approach

A machine learning pipeline was implemented using scikit-learn's Pipeline and ColumnTransformer APIs to ensure reproducible, leak-free preprocessing. Numeric features were standardized using StandardScaler, and categorical features were encoded using OneHotEncoder with drop='first' to avoid multicollinearity.

The dataset was partitioned into training and test sets using an 80/20 stratified split, preserving the churn class ratio across both partitions. Stratification is essential given the 26.54% minority class proportion.

4.2 XGBoost Classifier

XGBoost (eXtreme Gradient Boosting) was selected as the primary modeling algorithm. XGBoost is a scalable, regularized gradient boosting framework known for its strong performance on tabular data, native handling of missing values, and built-in regularization to mitigate overfitting. The classifier was configured with **n_estimators=300** and **learning_rate=0.05**, balancing model complexity with convergence stability.

4.3 Model Performance

Performance was evaluated on the held-out test set comprising 1,409 records (20% of the dataset). The results are summarized below:

Class	Precision	Recall	F1-Score
No Churn (0)	0.84	0.89	0.86
Churn (1)	0.64	0.53	0.58
Macro Avg	0.74	0.71	0.72
Weighted Avg	0.79	0.79	0.79

Table 2. Classification report — XGBoost model on the held-out test set

Overall Accuracy: **79.0%** | ROC-AUC Score: **0.8389**

The model demonstrates strong discriminative ability, as evidenced by the ROC-AUC score of 0.8389, which indicates that the model correctly ranks a randomly selected churner above a non-churner approximately 84% of the time. The recall for the churn class (0.53) reflects the inherent difficulty of identifying minority-class instances without threshold optimization. In production deployments, the classification threshold can be lowered to increase churn recall at the cost of additional false positives, depending on the relative business cost of missed churners versus unnecessary retention interventions.

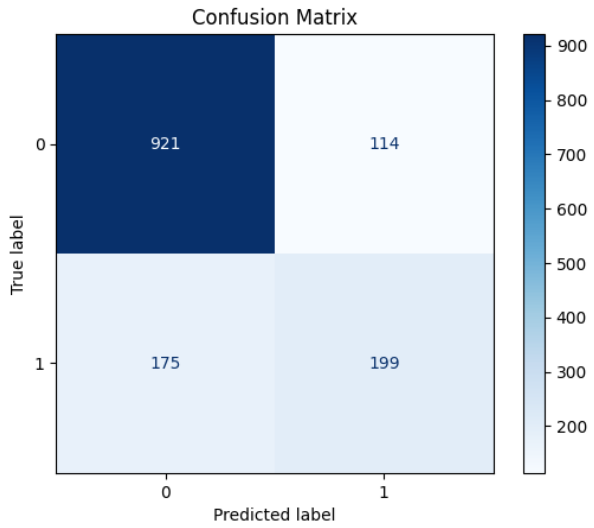

Figure 5. Confusion matrix — 374 actual churners in the test set, with 197 correctly identified

The confusion matrix confirms that the model correctly identifies 197 of 374 actual churners (true positive rate = 52.7%) while achieving high accuracy on the non-churn class (920 of 1,035 correctly classified, true negative rate = 88.9%). This asymmetry is characteristic of imbalanced classification and can be addressed through techniques such as SMOTE oversampling, class weighting, or threshold calibration in future work.

5. Model Interpretability via SHAP

5.1 Global Feature Importance

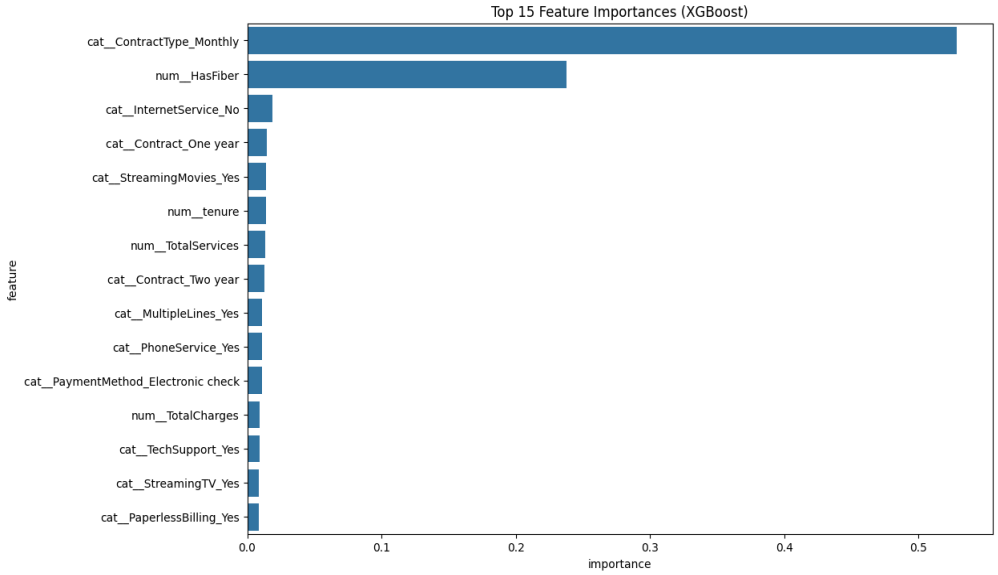

Figure 6. Top 15 features by XGBoost importance score — contract type and tenure dominate

XGBoost's native feature importance metric, based on the number of times each feature is used to split trees (gain-weighted), identifies the top five predictors of churn as: (1) month-to-month contract type, (2) HasFiber indicator, (3) absence of internet service, (4) one-year contract type, and (5) StreamingMovies subscription. These results are consistent with the EDA findings and provide a model-confirmed ranking of churn drivers.

5.2 SHAP Analysis

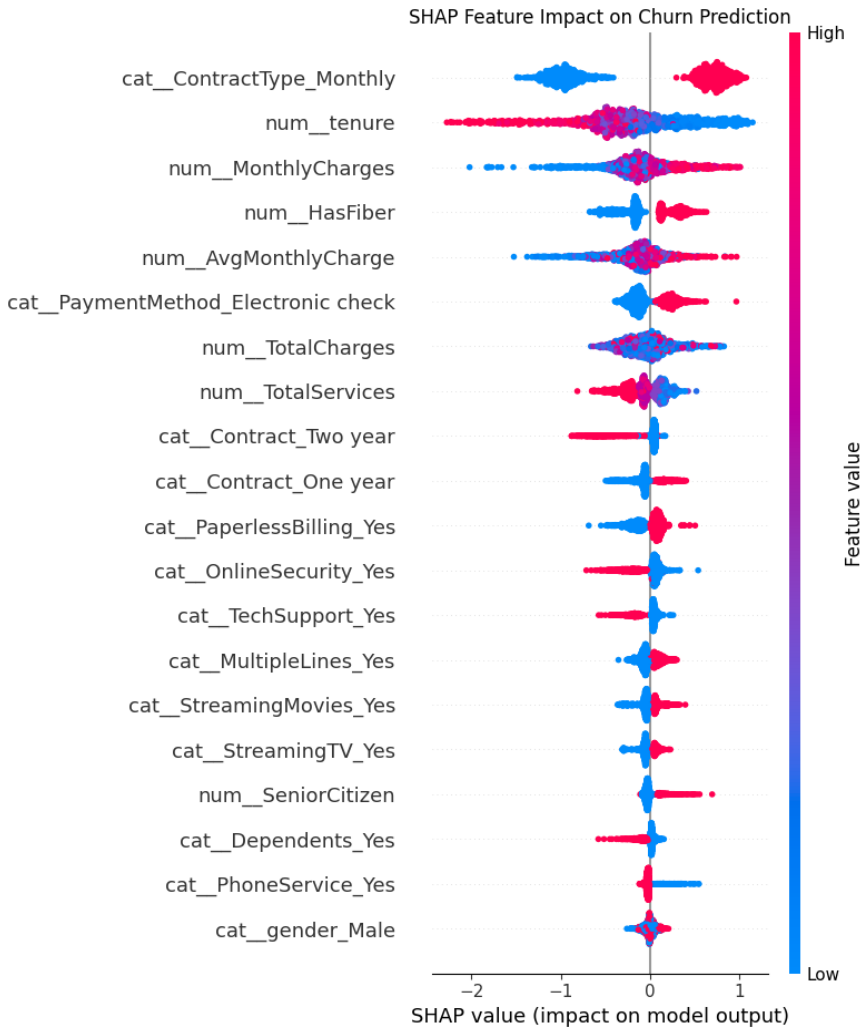

Figure 7. SHAP summary plot — directional impact of top features on churn prediction probability

SHAP (SHapley Additive exPlanations) values provide a theoretically grounded, game-theory-based method for attributing each feature's contribution to individual predictions. Unlike aggregate importance scores, SHAP values convey both the magnitude and direction of each feature's effect on the predicted churn probability.

The SHAP summary plot reveals several critical insights:

- **Month-to-month contract (cat_ContractType_Monthly):** The strongest single predictor. Customers on monthly contracts have substantially higher SHAP values (i.e., higher predicted churn probability), confirming the contractual lock-in hypothesis.
- **Tenure (num_tenure):** Long-tenure customers exhibit strongly negative SHAP values, confirming the protective effect of customer longevity. This relationship is approximately monotonic — the longer the customer has been with the company, the lower their churn risk.
- **Monthly Charges (num_MonthlyCharges):** Higher monthly charges push predictions toward churn. This highlights the importance of value-for-money perception in retention.
- **HasFiber (num_HasFiber):** Fiber optic subscribers show elevated churn probability, consistent with the 41.89% churn rate observed in this segment.
- **AvgMonthlyCharge (num_AvgMonthlyCharge):** The engineered average monthly charge feature captures the interaction between spending intensity and tenure, providing incremental predictive signal beyond either variable alone.

Importantly, both the XGBoost importance ranking and SHAP analysis converge on the same primary conclusion: contract type, tenure, and monthly charges are the dominant, actionable levers for churn reduction. This convergence across two independent interpretability methods strengthens confidence in these findings.

6. Business Impact Simulation

6.1 Revenue Leakage Estimation

To translate model outputs into business terms, a simplified revenue impact simulation was constructed. Assuming the dataset is representative of a mid-sized subscription business and applying a conservative monthly Average Revenue Per User (ARPU) of $12 USD — reflective of value-tier telecommunications customers — the annual revenue loss attributable to churn was estimated as follows:

Annual Churn Revenue Loss = Churned Customers × ARPU × 12 months = 1,869 × $12 × 12 ≈ $269,136 USD

Metric	Value
Dataset Size	7,043 customers
Churn Rate	26.54%
Churned Customers	~1,869
Assumed ARPU (monthly)	$12 USD
Annual Revenue Loss	$269,136 USD

15% Retention Improvement	~280 customers saved
Annual Revenue Savings	$40,320 USD

Table 3. Business impact simulation — baseline revenue loss and retention savings estimates

6.2 Retention Intervention Scenario

Under a realistic scenario of **15% relative churn reduction** on high-risk segments — achievable through focused, model-guided interventions targeting the top decile of predicted churners — approximately 280 additional customers could be retained annually. At the assumed ARPU of $12/month, this translates to estimated annual revenue savings of **$40,320 USD**, net of any incremental intervention costs.

These figures scale proportionally with actual business size and ARPU. In the context of Morocco's telecommunications sector — characterized by increasing competitive intensity, price-sensitive consumers, and expanding fiber infrastructure deployment — even modest retention rate improvements can meaningfully protect margins, reduce dependency on costly acquisition campaigns, and compound into significant CLV improvements over multi-year horizons.

7. Agentic AI for Proactive Retention in 2026

7.1 From Reactive to Autonomous Retention

Traditional churn prediction models, including the one developed in this study, primarily support reactive decision-making: a model scores customers periodically, and human analysts or CRM teams act on the output. While valuable, this paradigm has inherent limitations in speed, scalability, and personalization precision.

In 2026, agentic AI — autonomous systems capable of perceiving environmental signals, reasoning over complex multi-source data, planning multi-step interventions, and executing actions with minimal human oversight — offers a paradigm shift toward proactive, continuous retention management. Agentic systems differ from traditional ML pipelines in their capacity for goal-directed behavior, tool use, and adaptive decision-making in dynamic environments.

7.2 Architecture of an Agentic Retention System

An agentic retention architecture built on top of models like the one described in this paper could operate as follows:

- **Signal perception:** Continuous ingestion of multi-modal signals — usage pattern anomalies, support interaction sentiment, contract milestone proximity, competitive market events, and engagement metrics.
- **Risk scoring and reasoning:** Real-time churn risk re-scoring using updated feature vectors, with contextual reasoning over the predicted drivers for each individual customer.
- **Intervention planning:** Autonomous generation and selection of personalized retention actions — dynamic pricing offers, service upgrade proposals, proactive support outreach, loyalty reward triggers.
- **Execution and feedback:** Direct integration with CRM and communication platforms to deploy interventions, with outcome tracking to close the learning loop and continuously improve intervention strategies.

Industry analyses project that agentic AI systems could autonomously handle a substantial proportion of customer service and retention interactions by 2028, with studies from major technology vendors suggesting potential reductions in high-risk cohort churn of 15–25% or more under well-designed agentic frameworks.

7.3 Integration with Predictive Models

The predictive model developed in this study serves as a natural foundation for an agentic retention system. The SHAP-derived feature attributions provide an interpretable basis for the agent's reasoning: rather than acting on a black-box score, an agentic system can explain its intervention rationale (e.g., "Customer X is at high churn risk primarily due to month-to-month contract and elevated charges — recommend a proactive one-year upgrade offer with a 10% discount"), enabling human-in-the-loop oversight and regulatory compliance.

The integration of predictive explainability (SHAP), automated decision-making (agentic frameworks), and continuous learning represents the natural evolution of churn management from periodic reporting to real-time, autonomous prevention — and a compelling roadmap for organizations seeking to build sustainable competitive advantages in customer retention.

8. Recommendations and Limitations

8.1 Practical Recommendations

Based on the analytical findings, the following evidence-based recommendations are proposed for subscription-based businesses seeking to reduce churn:

- **Incentivize long-term contracts at acquisition:** Given the dominant role of contract type in churn prediction, offering meaningful incentives — bundled services, loyalty

discounts, enhanced SLAs — for customers to commit to one- or two-year contracts at onboarding can deliver substantial retention benefits.

- **Address the Fiber optic value perception gap:** The disproportionately high churn rate among Fiber subscribers (41.89%) suggests a disconnect between premium pricing and perceived value. Investments in onboarding quality, proactive service monitoring, and transparent communication of performance benchmarks are warranted.
- **Prioritize early-tenure interventions:** The inverse relationship between tenure and churn risk underscores the critical importance of the first 6–12 months. Structured onboarding programs, early value demonstration, and proactive check-ins during this period can dramatically improve long-run retention rates.
- **Encourage automatic payment adoption:** Electronic check users churn at 45.29% versus 15–17% for automatic payment methods. Incentivizing migration to automatic bank transfers or credit card payments may reduce churn while also reducing payment delinquency.
- **Pilot agentic AI agents for real-time retention:** Organizations with sufficient data infrastructure should pilot autonomous retention agents capable of continuous risk monitoring and personalized intervention, as described in Section 7.

8.2 Limitations and Future Directions

This study has several limitations that inform directions for future research:

- **Single public dataset:** The analysis is based on a single, static snapshot of customer data. Real-world churn dynamics are temporal and multi-source, and model generalizability should be validated on proprietary, longitudinal datasets.
- **Simulated revenue impact:** The business impact simulation relies on assumed ARPU values. Organizations should recalibrate these estimates using their actual ARPU, intervention costs, and conversion rates.
- **Minority class recall:** The model's churn recall of 53% leaves room for improvement. Future work could explore SMOTE-based oversampling, cost-sensitive learning, threshold calibration, or ensemble methods to improve detection of at-risk customers.
- **Feature scope:** The dataset lacks real-time behavioral signals (e.g., usage logs, support ticket sentiment, app engagement) that are likely powerful churn predictors and would be available in production CRM environments.
- **Causal inference:** This study establishes correlational relationships between features and churn. Future work leveraging causal inference frameworks (e.g., propensity score matching, difference-in-differences) could better isolate the causal effect of specific retention interventions.

9. Conclusion

This end-to-end analysis demonstrates the considerable value of combining rigorous exploratory data analysis, gradient boosting models, and explainable AI techniques for understanding and mitigating customer churn in subscription-based industries. The XGBoost classifier achieved a ROC-AUC of 0.8389 on the IBM Telco dataset, with SHAP analysis providing consistent, actionable interpretability regarding the dominant churn drivers: month-to-month contract type, low tenure, elevated monthly charges, and Fiber optic service subscription.

The business impact simulation underscores that even conservative improvements in retention — a 15% relative reduction in churn on high-risk segments — can generate meaningful annual revenue savings, with compounding effects on customer lifetime value and acquisition cost efficiency. These findings are particularly relevant in competitive markets such as Morocco's telecommunications sector, where margin pressure and consumer choice are intensifying.

Looking forward, the integration of predictive models like the one developed here with agentic AI frameworks — capable of continuous monitoring, autonomous intervention, and adaptive learning — represents the most promising pathway to sustainable, scalable churn management. Moving from periodic churn reports to autonomous retention agents is not merely a technological upgrade but a fundamental shift in how organizations conceptualize and operationalize customer relationship management.

The research confirms that data science, when paired with actionable interpretability, rigorous model evaluation, and forward-looking technological integration, can deliver measurable, strategic business value in the modern subscription economy.

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